

Exploring Fine-Tuning for Tabular Foundation Models

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Abstract

Tabular Foundation Models (TFMs) have recently shown strong in-context learning capabilities on structured data, achieving zero-shot performance comparable to traditional machine learning methods. We find that zero-shot TFMs already achieve strong performance, while the benefits of fine-tuning are highly model- and data-dependent. Meta-learning and PEFT provide moderate gains under specific conditions, whereas full supervised fine-tuning often reduces accuracy or calibration quality. This work¹ presents the first comprehensive study of fine-tuning in TFMs across benchmarks including TALENT, OpenML-CC18, and TabZilla. We compare zero-shot, meta-learning, supervised (SFT), and parameter-efficient (PEFT) approaches, analyzing how dataset factors such as imbalance, size, and dimensionality affect outcomes. Our findings cover performance, calibration, and fairness, offering practical guidelines on when fine-tuning is most beneficial and its limitations.

CCS Concepts

• **Computing methodologies** → *Machine learning algorithms; Learning paradigms.*

Keywords

fine-tuning, tabular data, foundation models, calibration, fairness

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1 Introduction

Tabular data drives applications in enterprises like healthcare, finance, and analytics. Although deep learning has transformed NLP and vision, classical models such as XGBoost, LightGBM, and CatBoost still dominate tabular prediction due to their robustness on heterogeneous, small, and medium-sized datasets. Recent *Tabular Foundation Models (TFMs)* including TabPFN [6, 7], TabICL [10], OrionMSP [3], OrionBiX [2], TabDPT [8], and Mitra [13] extend

the foundation-model paradigm to structured data, offering strong zero-shot performance and in-context learning. However, it remains unclear *when* fine-tuning helps these models, *which* strategies are effective, and *how* dataset properties influence outcomes. The wide variation in tabular dataset size, imbalance, and feature structure makes the impact of fine-tuning particularly non-obvious.

This work presents the first large-scale study of fine-tuning TFMs. Our contributions are: (I) a unified evaluation of four adaptation strategies: zero-shot, meta-learning, supervised fine-tuning (SFT), and parameter-efficient fine-tuning (PEFT) across six TFMs and multiple benchmarks; (II) an analysis of how dataset factors (size, imbalance, dimensionality) shape fine-tuning outcomes; (III) an assessment of calibration and fairness effects; and (IV) practical guidelines on when fine-tuning is effective. Our findings show that fine-tuning is not universally beneficial, underscoring the need for careful strategy selection.

2 Methodology

We evaluate six Tabular Foundation Models: TabPFN, TabICL, OrionMSP, OrionBiX, TabDPT, and Mitra alongside classical baselines (XGBoost, LightGBM, CatBoost, Random Forest).

Experiments span three benchmark suites: TALENT (155) [12], OPENML-CC18 (63) [1], and TABZILLA (27) [9]. For fairness evaluation, we additionally use nine public datasets: *Adult Census Income*, *German Credit (ID-31)*, *COMPAS Recidivism*, *Default of Credit Card Clients (Taiwan)*, *Student Performance (Portuguese)*, *Law School Admission*, *IBM HR Analytics Employee Attrition*, *Home Mortgage Disclosure Act (NY 2023)*, and *NHANES Demographics (2013–14)*. We follow standardized splits.

We performed our experiments on TFMs using the TabTune [11], and on baseline ML models using AutoGluon [4]. All experiments are run on NVIDIA L40S (H200 for memory-intensive settings).

Adaptation Strategies: We compare four paradigms. ZERO-SHOT INFERENCE evaluates pretrained TFMs without updating parameters. META-LEARNING FINE-TUNING uses episodic training with 48–512 support samples for 5 epochs. SUPERVISED FINE-TUNING (SFT) performs full-parameter optimization using AdamW with learning rates $1e^{-5}$ – $5e^{-5}$ and early stopping for up to 10 epochs. PARAMETER-EFFICIENT FINE-TUNING (PEFT) [5] employs LoRA with rank $r=8$ and $\alpha=16$, reducing memory usage; it is applied to both supervised fine-tuning and meta-learning variants.

Evaluation Metrics: Performance is measured using Accuracy, weighted F1, and mean rank across datasets. Calibration is assessed using Expected Calibration Error (ECE), Maximum Calibration Error (MCE), and Brier Score. Fairness is evaluated via Statistical Parity Difference (SPD), Equalized Odds Difference (EOD), and Equal Opportunity Difference (EOPD).

¹ Code for the experiments will be available at <https://github.com/Lexsi-Labs/TabTune/tree/main/experiments/WWW2026>



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Table 1: Overall leaderboard across three benchmark suites : TALENT, OpenML-CC18, and TabZilla. Ranks denote the mean rank per benchmark suite (lower is better). Metrics: ACC = Accuracy, F1 = Weighted F1. The "All" column reports the aggregated rank across for a strategy. Models are grouped by adaptation strategy. Formatting: 1st place; 2nd place within each group. *Note : that rank reflects only relative ordering and may not capture absolute ACC/F1 trends, which can vary across scenarios.*

Strategy	All	TALENT			OpenML-CC18			TabZilla		
Models	Rank(↓)	Rank(↓)	ACC(↑)	F1(↑)	Rank(↓)	ACC(↑)	F1(↑)	Rank(↓)	ACC(↑)	F1(↑)
<i>Baselines + Zero-Shot Inference</i>										
XGBoost	6.39	5.97	0.8403	0.8360	5.34	0.8558	0.8537	5.62	0.8612	0.8326
CatBoost	6.16	5.54	0.8336	0.8259	6.64	0.8588	0.8520	5.40	0.8579	0.8384
Random Forest	7.06	6.11	0.8285	0.8209	5.78	0.8547	0.8497	7.72	0.8358	0.8399
LightGBM	6.50	6.07	0.8331	0.8245	5.60	0.8581	0.8493	4.90	0.8618	0.8211
TabCL	4.78	4.07	0.8471	0.8372	4.36	0.8667	0.8623	5.63	0.8734	0.8698
OrionBiX	5.27	4.56	0.8346	0.8260	4.97	0.8653	0.8596	4.63	0.8728	0.8628
OrionMSP	3.76	3.26	0.8461	0.8360	3.90	0.8722	0.8676	3.78	0.8821	0.8786
TabPFN	4.44	3.71	0.8514	0.8412	4.36	0.8714	0.8663	4.65	0.8752	0.8716
Mitra	10.85	9.67	0.3921	0.2868	9.55	0.3614	0.2522	10.21	0.3152	0.1830
TabDPT	5.27	5.16	0.8408	0.8318	<u>4.32</u>	0.8672	0.8625	<u>3.81</u>	<u>0.8814</u>	<u>0.8775</u>
<i>FineTune - Meta Learning</i>										
TabCL	3.66	3.33	0.8344	0.8253	3.05	0.8664	0.8597	4.15	0.6956	0.6845
OrionBiX	3.96	4.25	0.8158	0.8060	2.86	<u>0.8548</u>	<u>0.8516</u>	<u>2.00</u>	<u>0.8726</u>	<u>0.8662</u>
OrionMSP	2.26	1.80	0.8401	0.8310	2.82	<u>0.8548</u>	<u>0.8516</u>	1.73	0.8735	0.8672
TabPFN	2.42	2.07	0.8517	0.8414	2.42	0.8842	0.8784	2.42	0.8663	0.8603
Mitra	6.09	5.79	0.6416	0.5874	5.42	0.6164	0.5651	4.70	0.5592	0.5147
TabDPT	3.95	3.72	0.8255	0.8167	4.40	0.8534	0.8501	2.62	0.8500	0.8456
<i>FineTune - Supervised Fine Tuning</i>										
TabCL	5.05	4.74	0.7668	0.7439	4.36	0.6838	0.6299	4.65	0.5670	0.4733
OrionBiX	4.21	3.54	0.7698	0.7469	4.30	0.7126	0.6595	4.25	0.6476	0.5782
OrionMSP	2.88	2.27	0.7908	0.7653	3.18	0.7995	0.7668	2.88	0.7454	0.7222
TabPFN	1.97	1.83	0.8459	0.8350	1.89	0.8697	0.8617	1.86	0.8433	0.8327
Mitra	5.98	5.62	0.5460	0.4382	5.02	0.5408	0.4309	5.52	0.4608	0.3467
TabDPT	<u>2.79</u>	2.98	0.8202	0.8094	<u>2.25</u>	<u>0.8499</u>	<u>0.8424</u>	1.81	<u>0.8337</u>	<u>0.8260</u>
<i>FineTune - PEFT - Meta Learning</i>										
TabCL	4.32	4.07	0.7017	0.6867	4.07	0.7773	0.7605	3.45	0.7116	0.7003
OrionBiX	2.77	2.39	0.7854	0.7762	2.50	0.8471	0.8430	3.52	0.7370	0.7200
OrionMSP	2.21	2.25	0.7879	0.7728	1.93	0.8566	0.8453	1.33	0.8594	0.8581
Mitra	4.64	4.17	0.6369	0.5905	4.32	0.6000	0.5426	4.36	0.5461	0.4960
TabDPT	<u>2.28</u>	2.11	0.8002	0.7910	2.17	0.8600	0.8539	2.33	<u>0.8495</u>	<u>0.8481</u>
<i>FineTune - PEFT - Supervised Fine Tuning</i>										
TabCL	3.70	3.72	0.5692	0.4638	3.06	0.8174	0.7965	2.92	0.7920	0.7776
OrionBiX	3.99	3.57	0.6358	0.5570	3.67	0.7380	0.6789	3.95	0.6707	0.6071
OrionMSP	2.01	2.01	0.6749	0.5956	1.73	0.8241	0.8033	1.47	0.8214	0.8071
Mitra	4.62	4.01	0.5294	0.4303	4.49	0.4965	0.3917	4.70	0.4606	0.3597
TabDPT	1.91	1.65	0.7988	0.7842	<u>2.03</u>	0.8500	0.8398	<u>1.95</u>	0.8461	0.8461

In this work, we study : Whether fine-tuning improves TFM performance; how dataset properties such as size, imbalance, and dimensionality influence gains; how adaptation strategies affect calibration; and the fairness implications of fine-tuning.

3 Experimental Results

We assess zero-shot and fine-tuned Tabular Foundation Models (TFMs) across performance, calibration, and fairness. Table 1 summarizes results on TALENT, OpenML-CC18, and TabZilla. Across accuracy and F1, zero-shot inference remains competitive and often outperforms fine-tuned variants, with several models showing reduced performance after SFT.

Across all suites, zero-shot TFMs already achieve strong accuracy and F1, often surpassing or matching their fine-tuned variants. OrionMSP and TabPFN provide the most competitive zero-shot results. Meta-learning offers moderate and architecture-dependent gains, particularly for TabPFN and OrionMSP, but full supervised fine-tuning (SFT) frequently reduces accuracy and F1 most notably for TabICL and OrionBiX. PEFT variants recover part of this lost performance and typically track meta-learning behaviour while retaining their computational efficiency, with TabDPT showing the most stable improvements under PEFT. Models' responses to adaptation vary, reflecting the distinct pretraining processes behind each architecture.

Performance Across Dataset Characteristics: Table 2 breaks down performance by dataset size, imbalance, and feature dimensionality. Fine-tuning is most beneficial on medium-sized datasets (1K–10K samples), where TabPFN and OrionMSP show small but consistent gains. On small datasets (<1K), zero-shot inference remains clearly superior, as most models overfit when fine-tuned. On large datasets (>10K), adaptation provides limited benefit beyond zero-shot predictions, with OrionMSP and TabDPT already performing strongly. For imbalanced datasets, meta-learning offers the most reliable improvements; SFT continues to degrade several architectures. Wide-feature datasets (>100 features) exhibit the few cases where fine-tuning provides noticeable boosts, such as for TabPFN (SFT). These mixed effects within each regime indicate that dataset heterogeneity, feature correlations, noise, and distributional skew strongly shapes how models respond to adaptation.

Calibration Evaluation : Zero-shot TFMs deliver the best overall calibration, with OrionMSP and TabPFN achieving the lowest ECE values across all benchmark suites as shown in Table 3. Meta-learning largely preserves this reliability while providing modest performance gains. In contrast, SFT substantially worsens calibration for most transformers, increasing both ECE and MCE; TabPFN is the only model that remains consistently well-calibrated after full fine-tuning. PEFT improves stability relative to SFT, but still does not match the calibration quality of zero-shot inference. Notably,

Table 2: Comprehensive performance analysis across dataset characteristics: size (Small <1K, Medium 1K–10K, Large >10K samples), class balance (Balanced ≥ 0.6 , Imbalanced <0.6), and feature dimensionality (Narrow <10, Medium 10–100, Wide >100 features). ACC = Accuracy; F1 = Weighted F1-score. Values are averaged across datasets within each category. Models are grouped by adaptation strategy. Formatting: 1st place; 2nd place within each group.

Strategy	Dataset Size						Dataset Imbalance				Dataset Width					
	Small (<1K)		Medium (1K–10K)		Large (>10K)		Balanced (≥ 0.6)		Imbalanced (<0.6)		Narrow (<10)		Medium (10–100)		Wide (>100)	
Models	ACC(↑)	F1(↑)	ACC(↑)	F1(↑)	ACC(↑)	F1(↑)	ACC(↑)	F1(↑)	ACC(↑)	F1(↑)	ACC(↑)	F1(↑)	ACC(↑)	F1(↑)	ACC(↑)	F1(↑)
<i>Baselines + Zero-Shot Inference</i>																
XGBoost	0.8168	0.7964	0.8363	0.8314	0.8969	0.8920	0.8175	0.8110	0.8859	0.8785	0.8222	0.8159	0.8482	0.8410	0.9140	0.9039
CatBoost	0.8124	0.7935	0.8340	0.8264	0.8797	0.8733	0.8076	0.8020	0.8785	0.8665	0.8145	0.8067	0.8441	0.8344	0.9157	0.9084
Random Forest	0.7988	0.8187	0.8285	0.8221	0.8694	0.8628	0.7983	0.7955	0.8741	0.8646	0.8005	0.7044	0.8410	0.8235	0.9034	0.8936
LightGBM	0.8143	0.7789	0.8314	0.8226	0.8827	0.8764	0.8071	0.7977	0.8775	0.8633	0.8128	0.7907	0.8458	0.8326	0.8999	0.8908
TabICL	0.8301	0.8338	0.8486	0.8398	0.8802	0.8743	0.8279	0.8233	0.8806	0.8698	0.8208	0.8119	0.8627	0.8549	0.9101	0.8936
OrionBiX	0.8330	0.8150	0.8348	0.8260	0.8729	0.8670	0.8096	0.8040	0.8787	0.8683	0.8112	0.8043	0.8510	0.8417	0.8859	0.8849
OrionMSP	0.8232	0.8194	0.8494	0.8402	0.8843	0.8768	0.8265	0.8202	0.8840	0.8731	0.8394	0.8314	0.8572	0.8478	0.8860	0.8837
TabPFN	0.8325	0.8131	0.8557	0.8462	0.8783	0.8713	0.8367	0.8309	0.8808	0.8697	0.8187	0.8092	0.8676	0.8589	0.9129	0.9111
Mitra	0.4334	0.3236	0.3600	0.2553	0.3837	0.2754	0.2763	0.1540	0.4794	0.3858	0.3737	0.2683	0.3886	0.2781	0.2521	0.1497
TabDPT	0.8333	0.8271	0.8424	0.8339	0.8831	0.8765	0.8233	0.8189	0.8798	0.8690	0.8262	0.8189	0.8566	0.8483	0.8845	0.8820
<i>FineTune – Meta Learning</i>																
TabICL	0.7912	0.7806	0.8342	0.8256	0.8431	0.8360	0.7943	0.7884	0.8662	0.8547	0.7867	0.7779	0.8383	0.8338	0.8319	0.8667
OrionBiX	0.8149	0.8123	0.8283	0.8204	0.8410	0.8306	0.7947	0.7881	0.8622	0.8532	0.7956	0.7900	0.8316	0.8278	0.8429	0.8640
OrionMSP	0.8211	0.8151	0.8413	0.8330	0.8787	0.8721	0.8231	0.8183	0.8735	0.8636	0.8253	0.8190	0.8514	0.8457	0.8564	0.8737
TabPFN	0.8336	0.8256	0.8638	0.8448	0.8748	0.8668	0.8461	0.8407	0.8784	0.8664	0.8264	0.8171	0.8683	0.8579	0.9172	0.9533
Mitra	0.6417	0.5988	0.6083	0.5516	0.6585	0.6132	0.4699	0.4142	0.7885	0.7391	0.6646	0.6247	0.6004	0.5763	0.4090	0.3224
TabDPT	0.8224	0.8188	0.8289	0.8212	0.8682	0.8613	0.8066	0.8023	0.8650	0.8552	0.8129	0.8071	0.8405	0.8350	0.8400	0.8497
<i>FineTune – Supervised FineTuning</i>																
TabICL	0.6746	0.6225	0.7553	0.7187	0.6693	0.6212	0.6708	0.6317	0.7794	0.7362	0.6609	0.6195	0.7507	0.7101	0.7757	0.7269
OrionBiX	0.7158	0.6792	0.7718	0.7386	0.6697	0.6272	0.6917	0.6550	0.7934	0.7590	0.6922	0.6576	0.7643	0.7294	0.7560	0.7046
OrionMSP	0.8135	0.7920	0.8036	0.7798	0.7231	0.6828	0.7467	0.7287	0.8307	0.7953	0.7540	0.7297	0.8025	0.7750	0.8148	0.7808
TabPFN	0.8216	0.8109	0.8580	0.8485	0.8572	0.8459	0.8336	0.8267	0.8709	0.8578	0.8258	0.8162	0.8578	0.8469	0.9346	0.9335
Mitra	0.5625	0.4408	0.5133	0.4116	0.5720	0.4501	0.3216	0.2037	0.7413	0.6415	0.5619	0.4478	0.5349	0.4268	0.3307	0.2436
TabDPT	0.8227	0.8153	0.8377	0.8284	0.8120	0.8002	0.8025	0.7949	0.8578	0.8464	0.8078	0.7948	0.8378	0.8294	0.8623	0.8587
<i>FineTune – PEFT – Meta Learning</i>																
TabICL	0.7518	0.7397	0.7251	0.7114	0.6962	0.6799	0.6530	0.6460	0.8026	0.7800	0.6375	0.6375	0.7601	0.7444	0.7714	0.7622
OrionBiX	0.8093	0.8042	0.7995	0.7939	0.7848	0.7665	0.7504	0.7436	0.8528	0.8420	0.7584	0.7584	0.8147	0.8057	0.8251	0.8243
OrionMSP	0.8275	0.8213	0.8134	0.8019	0.8013	0.7817	0.7714	0.7623	0.8608	0.8433	0.7836	0.7836	0.8254	0.8107	0.8293	0.8256
Mitra	0.6510	0.6012	0.6089	0.5599	0.6304	0.5795	0.4683	0.4173	0.8001	0.7524	0.6471	0.6471	0.6177	0.5650	0.4129	0.3338
TabDPT	0.8191	0.8157	0.8121	0.8049	0.8407	0.8299	0.7860	0.7801	0.8609	0.8511	0.7979	0.7979	0.8299	0.8209	0.8416	0.8376
<i>FineTune – PEFT – Supervised FineTuning</i>																
TabICL	0.7957	0.7710	0.6424	0.5686	0.6084	0.5020	0.6003	0.5233	0.7223	0.6507	0.6196	0.5287	0.6596	0.5902	0.8760	0.8502
OrionBiX	0.7576	0.7102	0.6585	0.5892	0.6327	0.5381	0.5974	0.5164	0.7490	0.6875	0.6278	0.5444	0.6851	0.6177	0.6737	0.6166
OrionMSP	0.8298	0.8132	0.7190	0.6597	0.6931	0.6152	0.6753	0.6105	0.7915	0.7423	0.6795	0.6113	0.7477	0.6939	0.8042	0.7663
Mitra	0.5244	0.4133	0.5008	0.4013	0.5459	0.4476	0.3189	0.2171	0.7495	0.6497	0.5518	0.4481	0.5106	0.4108	0.2831	0.1874
TabDPT	0.8269	0.8182	0.8332	0.8245	0.7686	0.7418	0.7907	0.7798	0.8496	0.8346	0.7821	0.7631	0.8305	0.8201	0.8925	0.8901

accuracy and calibration respond differently to adaptation, with accuracy gains rarely translating into improved confidence.

Fairness Evaluation : Fairness results in Table 4 reveal a clear accuracy–equity trade-off. Mitra achieves the lowest disparity metrics but at significantly reduced predictive performance. Zero-shot and meta-learning generally offer the best balance of accuracy and fairness, with OrionMSP and OrionBiX maintaining competitive accuracy and moderate disparity levels. SFT exhibits the largest swings in fairness, especially for TabICL and OrionBiX, whereas TabPFN remains comparatively stable across strategies. PEFT variants show mild fairness improvements in some settings but do not consistently outperform zero-shot or meta-learning.

Key Findings : Fine-tuning effectiveness is highly model- and data-dependent. TabPFN benefits the most from fine-tuning particularly SFT on medium and wide datasets, while remaining well-calibrated. OrionMSP excels primarily in zero-shot and meta-learning regimes, especially on large or imbalanced datasets. TabDPT achieves strong efficiency–performance trade-offs under PEFT. In contrast, TabICL consistently degrades under SFT, particularly on small or heterogeneous datasets. Overall, fine-tuning yields limited improvements and is most useful in mid-sized or high-dimensional settings. Zero-shot inference remains preferable in low-data regimes and when calibration or fairness stability is important.

3.1 Practical Guidelines :

- **PREFER ZERO-SHOT FOR SMALL DATA:** Zero-shot inference is strongest on small datasets (<1K samples) where fine-tuning often overfits.
- **USE FINE-TUNING SELECTIVELY:** Apply it mainly on medium-sized (1K–10K) or wide-feature datasets, where TabPFN and TabDPT show consistent benefits.
- **META-LEARNING AS DEFAULT:** Meta-learning provides modest but reliable gains and preserves calibration, particularly on imbalanced datasets.
- **CAUTION WITH SFT:** Full supervised fine-tuning frequently reduces accuracy and calibration and should be reserved for models tolerant to full updates (e.g., TabPFN).
- **PEFT FOR EFFICIENCY:** PEFT avoids overfitting and recovers most useful improvements, making it especially effective for TabDPT and OrionMSP.

Overall, aligning the adaptation strategy with dataset size, feature dimensionality, and calibration requirements is crucial, as zero-shot inference often remains competitive.

Limitations : Our evaluation focuses on binary and multi-class classification; PEFT is not supported for some models. Sensitive attributes for fairness are manually defined and vary across datasets.

Table 3: Probability calibration metrics across models and tuning strategies on TALENT, OpenML-CC18, and TabZilla benchmark suites. Values are averaged across datasets within each suite. Models are grouped by adaptation strategy.

Strategy	TALENT			OpenML-CC18			TabZilla		
Models	ECE(↓)	MCE(↓)	BS(↓)	ECE(↓)	MCE(↓)	BS(↓)	ECE(↓)	MCE(↓)	BS(↓)
<i>Zero-Shot Inference</i>									
TabICL	0.0219	0.2421	0.1533	0.0371	0.3404	0.1267	0.0369	0.2863	0.1301
OrionBiX	0.0324	0.2245	0.1787	0.0325	0.3230	0.1325	0.0385	0.2512	0.2419
OrionMSP	0.0219	0.2098	0.1589	0.0319	0.2902	0.1262	0.0310	0.2805	0.1243
TabPFN	0.0276	0.2470	0.1514	0.0375	0.2880	0.1253	0.0431	0.2745	0.1283
Mitra	0.2238	0.3115	0.5291	0.2138	0.2952	0.5307	0.1946	0.2995	0.5733
TabDPT	0.0308	0.2489	0.1586	0.0443	0.3160	0.1351	0.0435	0.2997	0.1227
<i>FineTune - Meta Learning</i>									
TabICL	0.0355	0.2570	0.1671	0.0426	0.3387	0.1346	0.2033	0.4926	0.3732
OrionBiX	0.0409	0.2693	0.1841	0.0731	0.4164	0.1607	0.0744	0.3964	0.1416
OrionMSP	0.0340	0.2570	0.1656	0.0520	0.3458	0.1606	0.0365	0.2574	0.1250
TabPFN	0.0271	0.2440	0.1502	0.0346	0.2989	0.1178	0.0388	0.2636	0.1344
Mitra	0.0334	0.1912	0.3479	0.0533	0.2258	0.3932	0.0891	0.2086	0.4239
TabDPT	0.0562	0.3235	0.1791	0.0745	0.4178	0.1620	0.0664	0.2838	0.1591
<i>FineTune - Supervised FineTuning</i>									
TabICL	0.0768	0.3219	0.2349	0.1156	0.3610	0.3039	0.1534	0.3280	0.4078
OrionBiX	0.0753	0.3204	0.2334	0.1131	0.3374	0.2923	0.1581	0.3801	0.2530
OrionMSP	0.0471	0.1886	0.2113	0.0550	0.3155	0.1995	0.0773	0.3321	0.2590
TabPFN	0.0287	0.2509	0.1580	0.0381	0.2957	0.1246	0.0469	0.2528	0.1454
Mitra	0.1723	0.2620	0.4677	0.1539	0.2269	0.4784	0.2023	0.2991	0.5254
TabDPT	0.0548	0.3019	0.1867	0.0497	0.3397	0.1443	0.0713	0.3183	0.1801
<i>PEFT FineTune - Meta Learning</i>									
TabICL	0.0355	0.2570	0.1671	0.0426	0.3387	0.1346	0.2033	0.4926	0.3732
OrionBiX	0.0409	0.2693	0.1841	0.0731	0.4164	0.1607	0.0744	0.3964	0.1416
OrionMSP	0.0340	0.2570	0.1656	0.0520	0.3458	0.1606	0.0365	0.2574	0.1250
TabPFN	0.0334	0.1912	0.3479	0.0533	0.2258	0.3932	0.0891	0.2086	0.4239
TabDPT	0.0562	0.3235	0.1791	0.0745	0.4178	0.1620	0.0664	0.2838	0.1591
<i>PEFT FineTune - Supervised FineTuning</i>									
TabICL	0.0768	0.3219	0.2349	0.1156	0.3610	0.3039	0.1534	0.3280	0.4078
OrionBiX	0.0753	0.3204	0.2334	0.1131	0.3374	0.2923	0.1581	0.3801	0.2530
OrionMSP	0.0471	0.1886	0.2113	0.0550	0.3155	0.1995	0.0773	0.3321	0.2590
TabPFN	0.0287	0.2509	0.1580	0.0381	0.2957	0.1246	0.0469	0.2528	0.1454
Mitra	0.1723	0.2620	0.4677	0.1539	0.2269	0.4784	0.2023	0.2991	0.5254
TabDPT	0.0548	0.3019	0.1867	0.0497	0.3397	0.1443	0.0713	0.3183	0.1801

Table 4: Fairness metrics across models and tuning strategies on datasets with explicit demographic attributes. Values are averaged across fairness evaluation datasets. Models are grouped by adaptation strategy.

Strategy	Fairness Metrics			Performance Metrics	
Models	SPD(↓)	EOD(↓)	EOpD(↓)	ACC(↑)	F1(↑)
<i>Zero-Shot Inference</i>					
TabICL	0.2900	0.3168	0.3278	0.8743	0.8680
OrionBiX	0.3328	0.2698	0.2855	0.8779	0.8727
OrionMSP	0.3380	0.2827	0.2983	0.8752	0.8706
TabPFN	0.3070	0.3114	0.3176	0.8708	0.8631
Mitra	0.0193	0.0590	0.0982	0.6902	0.5847
TabDPT	0.2990	0.3197	0.3338	0.8674	0.8582
<i>FineTune - Meta Learning</i>					
TabICL	0.3140	0.3100	0.3332	0.8678	0.8603
OrionBiX	0.2021	0.1624	0.2010	0.8743	0.8692
OrionMSP	0.3038	0.2798	0.2907	0.8680	0.8650
TabPFN	0.3070	0.3115	0.3177	0.8733	0.8668
Mitra	0.2586	0.2748	0.3301	0.7967	0.7546
TabDPT	0.2648	0.3180	0.3080	0.8614	0.8503
<i>FineTune - Supervised FineTuning</i>					
TabICL	0.0550	0.3225	0.3319	0.4584	0.3625
OrionBiX	0.1128	0.1761	0.1817	0.6180	0.5498
OrionMSP	0.1227	0.2208	0.2353	0.4753	0.4039
TabPFN	0.3070	0.3115	0.3177	0.8733	0.8668
Mitra	0.0161	0.0170	0.0555	0.7268	0.6277
TabDPT	0.3075	0.3153	0.3284	0.8529	0.8435
<i>PEFT FineTune - Meta Learning</i>					
TabICL	0.3140	0.3092	0.3332	0.8678	0.8602
OrionBiX	0.3149	0.2722	0.2875	0.8748	0.8717
OrionMSP	0.3038	0.2798	0.2907	0.867	0.8632
Mitra	0.2595	0.2391	0.3154	0.8104	0.7721
TabDPT	0.2778	0.2792	0.3051	0.8651	0.8556
<i>PEFT FineTune - Supervised FineTuning</i>					
TabICL	0.1572	0.4041	0.4089	0.4881	0.4269
OrionBiX	0.0349	0.1564	0.2234	0.6401	0.5759
OrionMSP	0.1132	0.2140	0.2420	0.5732	0.5080
Mitra	0.0029	0.0370	0.0740	0.7133	0.6028
TabDPT	0.2960	0.3190	0.3342	0.8395	0.8505

For consistency, results are reported only on the common subset available to all models within each benchmark suite: 155/181 for TALENT, 27/36 for TabZilla, and 63/72 for OpenML-CC18.

4 Conclusion

This work demonstrates that tabular foundation models achieve strong zero-shot performance, but fine-tuning provides selective benefits that depend heavily on model architecture and dataset characteristics. Although meta-learning and PEFT offer moderate gains in specific scenarios, full supervised fine-tuning often degrades performance and calibration. Our guidelines can help practitioners navigate these trade-offs, emphasizing the importance of understanding when adaptation helps versus when zero-shot suffices.

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